

GVASYUGI, Russian Federation

WMO: 318010

Lat: 47.6704N

Lon: 136.1836E

Elev: 591

StdP: 14.39

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-34.0	-30.3	-39.0	0.6	-33.6	-35.3	0.8	-30.2	12.2	-1.3	11.2	0.4	0.2	50	0.529

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.8	86.3	70.9	83.1	69.3	80.2	68.0	73.7	81.9	71.8	79.4	70.2	77.4	3.2	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.1	117.3	77.8	69.3	110.2	75.9	67.7	104.1	74.2	37.9	82.1	36.1	79.6	34.7	77.5	83.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
11.7	10.0	7.6	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-39.4	91.6	4.2	2.7	-42.5	93.5	-44.9	95.1	-47.3	96.6	-50.4	98.6		
			-39.5	78.0	4.2	2.6	-42.6	79.9	-45.0	81.4	-47.4	82.9	-50.4	84.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	33.6	-7.6	-0.3	17.5	36.8	52.1	62.4	69.5	67.0	55.4	38.2	14.8	-4.4
	DBStd	28.62	9.14	9.57	10.89	8.06	6.62	5.97	4.54	4.95	6.98	8.08	12.04	9.90
	HDD50	7802	1786	1408	1008	400	55	1	0	0	29	373	1055	1687
	HDD65	11746	2251	1828	1473	845	401	117	14	35	294	831	1505	2152
	CDD50	1829	0	0	0	5	120	374	604	528	191	7	0	0
	CDD65	298	0	0	0	0	1	39	153	98	7	0	0	0
	CDH74	2643	0	0	0	11	151	586	1219	615	59	2	0	0
	CDH80	678	0	0	0	1	28	167	368	109	5	0	0	0
Wind	WSAvg	2.2	1.7	2.3	2.9	2.8	2.9	2.1	1.7	1.7	2.0	2.4	2.4	1.8
Precipitation	PrecAvg	34.1	0.7	0.7	1.3	1.9	3.1	3.9	6.5	6.3	4.3	2.6	1.7	1.2
	PrecMax	54.4	1.3	1.4	2.7	4.5	7.1	6.9	12.8	11.8	12.3	4.7	2.6	3.3
	PrecMin	23.9	0.2	0.0	0.1	0.5	1.0	1.5	1.8	1.9	1.9	0.5	0.4	0.4
	PrecStd	7.2	0.3	0.4	0.8	1.0	1.3	1.5	2.6	2.4	2.1	1.1	0.7	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.6	34.1	50.9	74.0	83.3	89.6	90.7	86.8	79.7	69.4	47.9	28.4
		MCWB	19.3	28.7	38.8	52.8	60.8	69.1	72.8	73.0	66.8	55.0	40.8	25.6
	2%	DB	15.6	27.3	44.5	65.8	78.3	84.7	87.4	83.0	74.9	61.3	40.8	22.2
		MCWB	13.9	23.2	35.3	48.7	58.5	67.9	72.5	71.2	63.7	50.1	35.3	20.3
	5%	DB	11.5	22.3	39.7	58.4	73.7	80.7	84.3	80.3	71.3	55.9	35.9	16.7
		MCWB	10.0	19.3	32.2	44.8	55.8	65.6	70.7	70.0	62.1	47.3	32.5	15.1
	10%	DB	8.0	18.0	35.4	52.9	68.8	76.6	81.0	77.5	67.4	51.3	32.3	11.9
		MCWB	6.7	15.3	30.2	42.2	53.6	64.0	69.9	68.4	59.8	45.2	30.0	10.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.3	30.0	39.6	54.7	63.5	74.3	76.6	75.6	69.6	56.0	42.0	26.3
		MCDB	21.1	34.1	48.8	72.2	78.1	83.2	85.7	82.8	77.1	65.8	47.5	28.2
	2%	WB	13.8	23.5	36.1	49.9	60.2	69.6	74.3	73.2	66.0	52.0	36.3	20.4
		MCDB	15.3	27.0	42.8	62.4	73.5	81.0	83.4	79.8	72.3	58.5	39.8	22.2
	5%	WB	10.1	19.4	33.1	45.9	57.8	67.3	72.6	71.5	63.4	49.1	33.0	15.1
		MCDB	11.5	22.5	38.4	57.1	69.6	77.3	81.1	77.6	68.9	54.6	35.7	16.5
	10%	WB	6.7	15.3	30.4	42.6	55.6	65.4	70.9	69.8	61.1	45.8	29.8	10.5
		MCDB	7.9	18.0	35.5	51.9	65.2	74.0	78.6	76.0	66.2	50.5	32.4	11.8
Mean Daily Temperature Range	5% DB	MDBR	26.3	31.4	27.8	23.9	26.2	23.1	19.8	17.7	21.4	21.8	20.9	23.5
		MCDBR	27.6	32.9	30.1	37.0	37.7	32.0	27.2	22.5	26.7	31.1	23.1	24.0
		MCWBR	25.5	28.6	21.0	20.8	18.8	16.3	13.5	12.1	15.5	19.3	17.5	21.9
	5% WB	MCDBR	26.6	31.8	26.1	34.5	31.2	27.0	23.4	19.4	22.5	27.5	21.6	22.9
		MCWBR	24.7	27.8	18.4	20.1	16.7	14.8	13.2	11.6	14.0	18.4	16.6	21.1
Clear-Sky Solar Irradiance	taub	0.298	0.351	0.426	0.531	0.559	0.522	0.496	0.454	0.417	0.430	0.383	0.316	
	taud	2.062	1.904	1.768	1.729	1.774	1.962	2.089	2.213	2.221	2.014	2.008	2.022	
	Ebn at Noon	235	243	242	229	229	240	245	251	248	214	195	207	
	Edh at Noon	30	43	59	68	68	56	49	42	37	39	32	27	
All-Sky Solar Radiation	RadAvg	609	982	1351	1411	1609	1698	1615	1420	1172	786	515	453	
	RadStd	33	42	108	141	134	219	147	98	97	69	47	26	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (7 neighbors)	+0.45	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page